



Progress Towards the Facility for Rare Isotope Beams

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On Behalf of FRIB Accelerator Team & Collaboration

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MICHIGAN STATE
UNIVERSITY



U.S. DEPARTMENT OF
ENERGY

Office of
Science

Outline

- Introduction
- Design philosophy
- Accelerator physics challenges
- Technology challenges
- Acquisition
- Future perspectives
- Acknowledgements



FRIB on Accelerator Beam-power Frontier

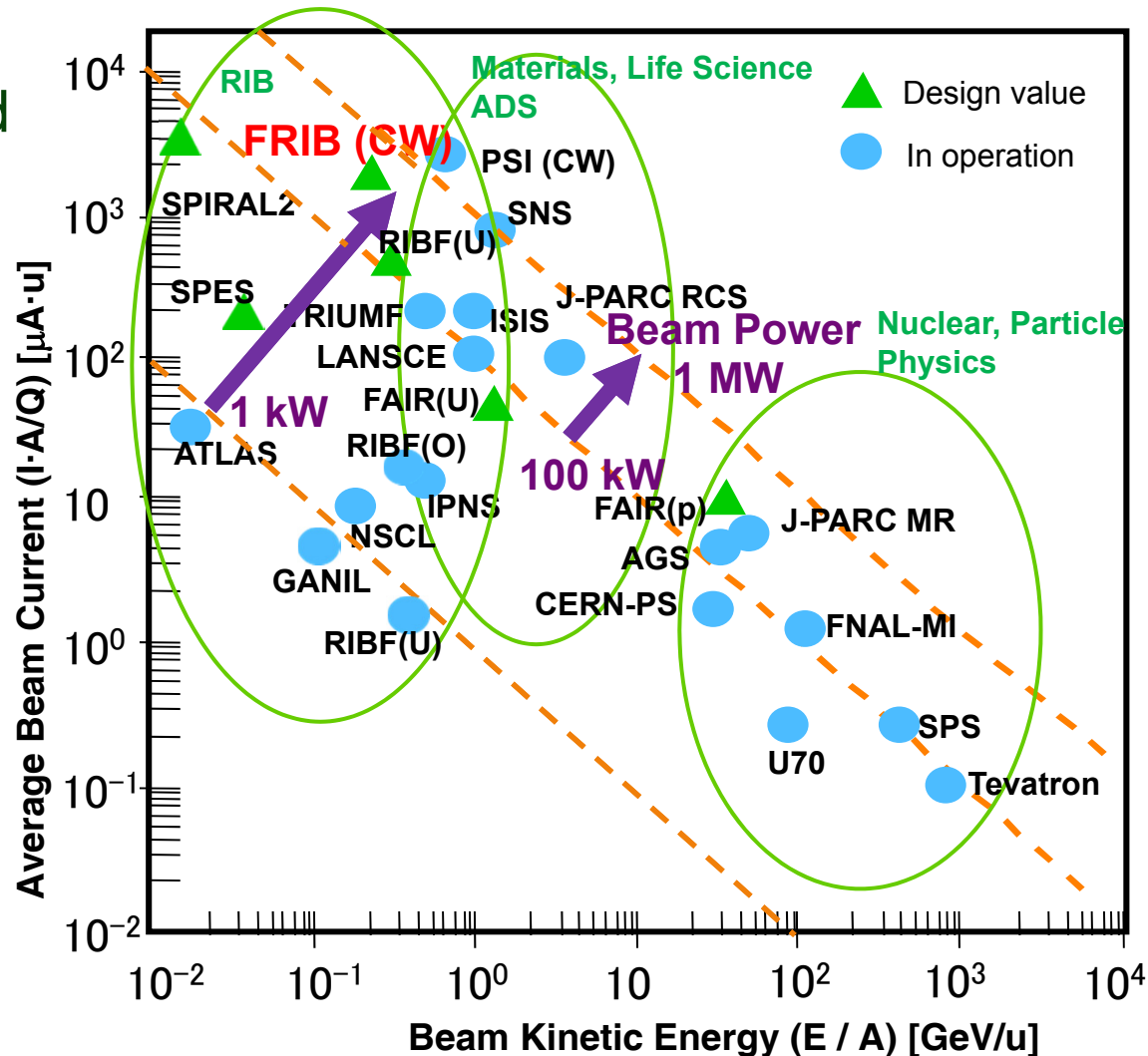
Raising Heavy-ion Beam Power by Two Orders-of-Magnitude

- During the past decade, proton accelerators raised beam power by x10

- SNS (USA): 1 MW pulsed; SRF linac/accumulator
- J-PARC (Japan): 0.3 MW pulsed; warm linac/RCS
- PSI (Switzerland): 1.4 MW CW; cyclotron

- FRIB will raise heavy-ion beam power from ~1 kW to 400 kW

- From proton to ^{238}U
- Using SRF linac from 0.5 MeV/u to > 200 MeV/u



Facility for Rare Isotope Beams
U.S. Department of Energy Office of Science
Michigan State University

FRIB Project at Michigan State University

Project of \$730M (\$635.5M DOE, \$94.5M MSU)

- 2008-12: DOE selects MSU to establish FRIB
- 2009-6: DOE and MSU sign corresponding cooperative agreement
- 2010-9 CD-1: conceptual design complete & preferred alternatives decided
- 2013-8 CD-2/CD-3a: performance baseline, start of civil construction & long lead procurement

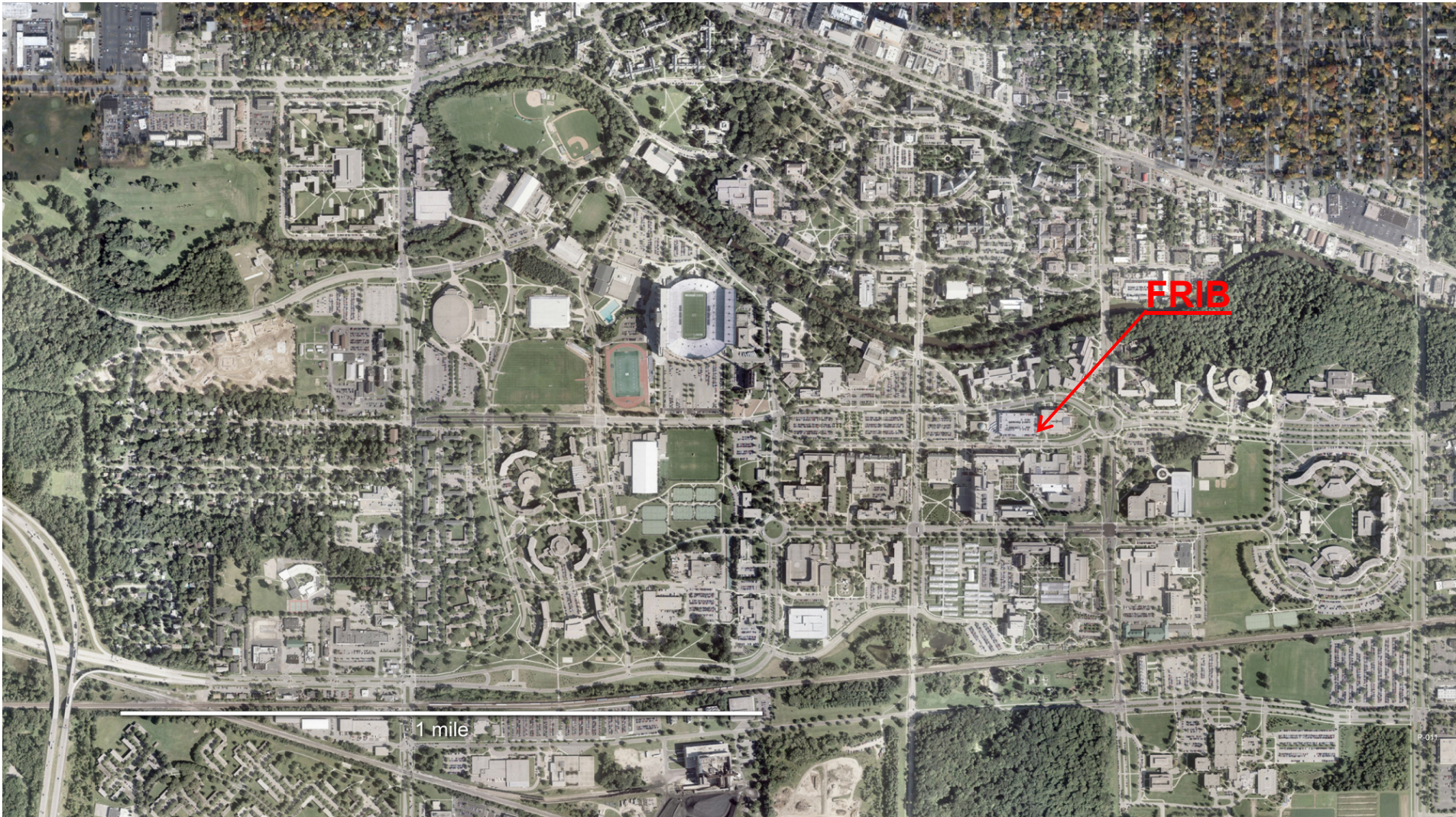
Growth from more than 500 employees today at NSCL, MSU

More than 1200 registered user at NSCL user group and at FRIB user organization

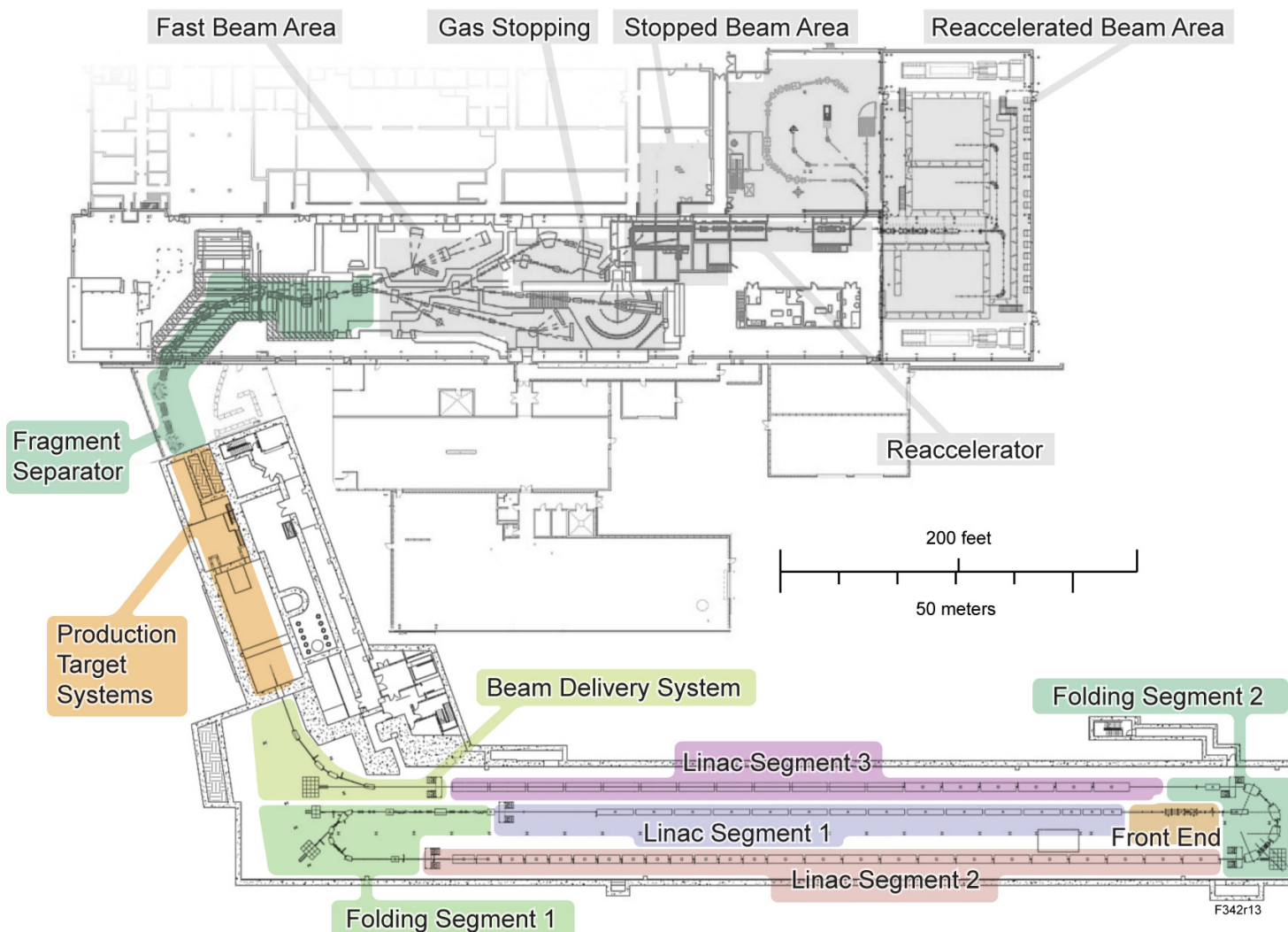


Michigan State University

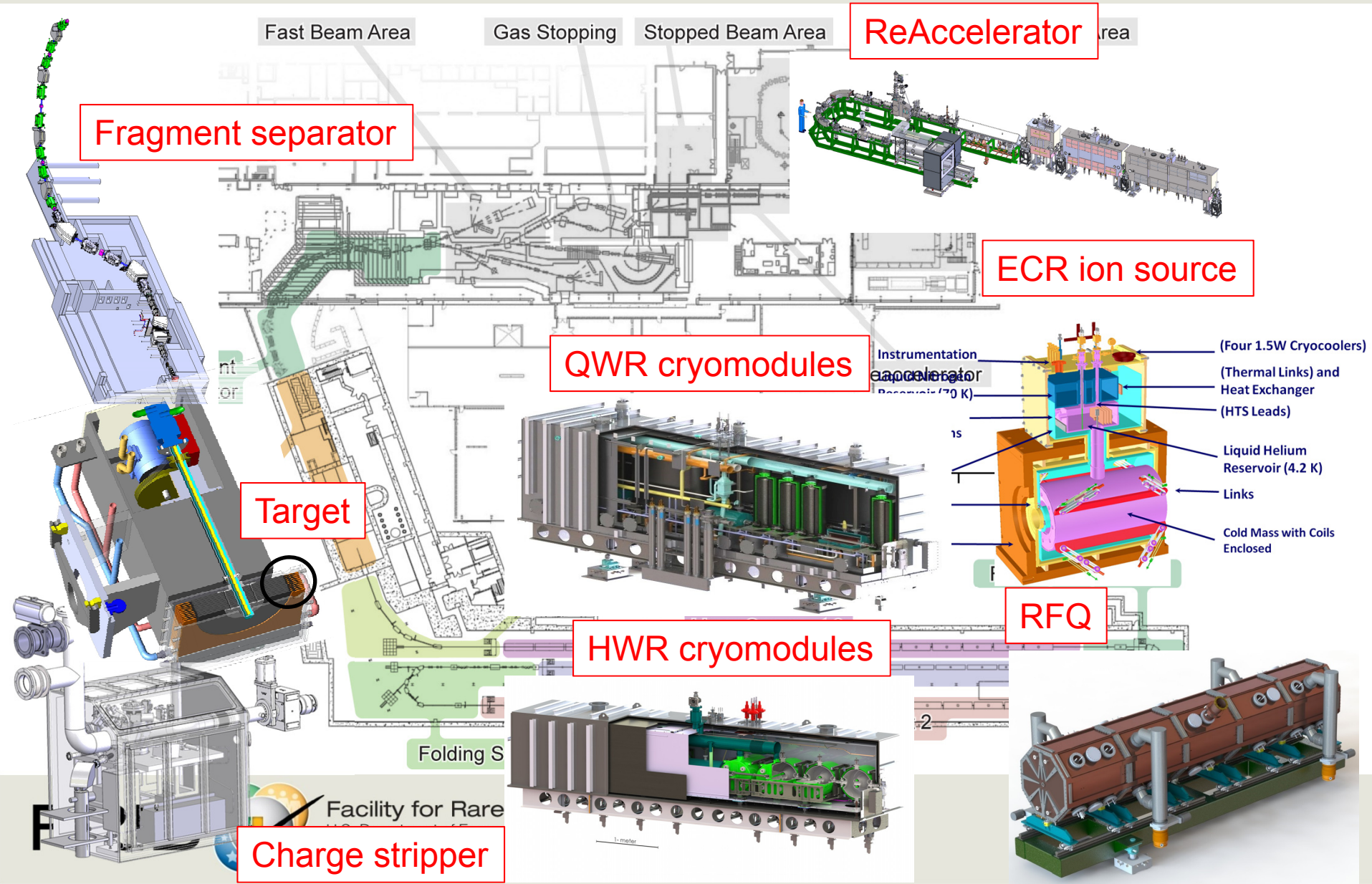
57,000 people; 36 sq mi; \$1.8B annual revenue; 552 buildings



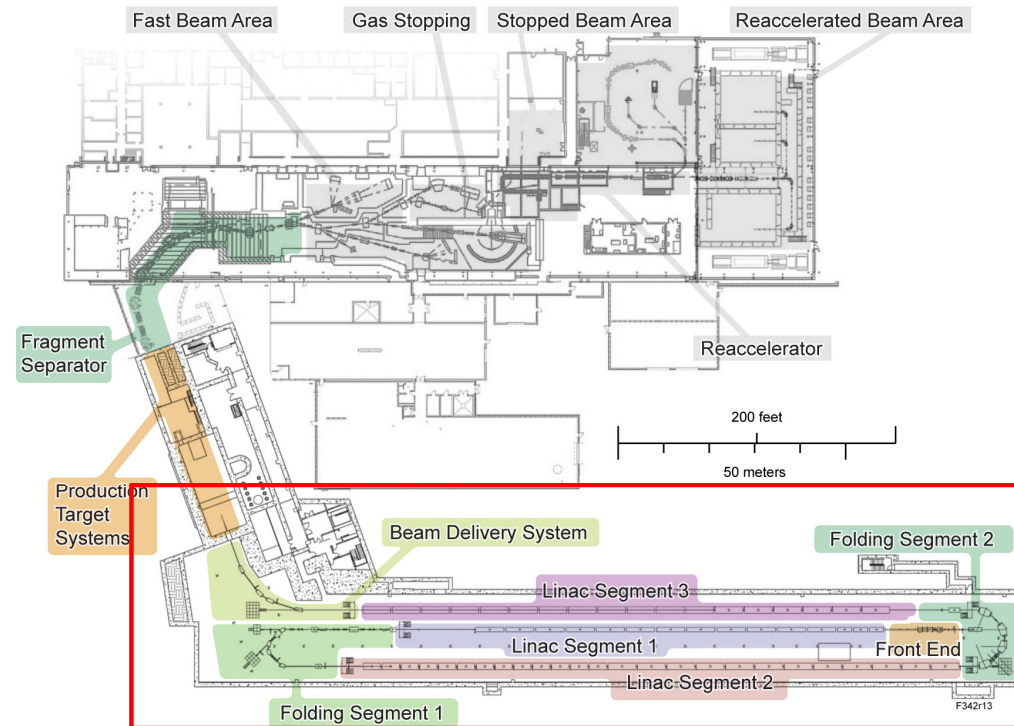
FRIB Accelerator Complex Subsystems



FRIB Accelerator Complex Subsystems



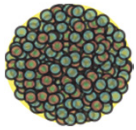
FRIB Accelerator Design Requirements



- Delivers FRIB accelerator as part of a DOE-SC national user facility with high reliability & availability
- Accelerate ion species up to ^{238}U with energies of no less than 200 MeV/u
- Provide beam power up to 400 kW
- Satisfy beam-on-target requirements

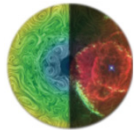
- Energy upgrade by filling vacant slots with 12 SRF cryomodules
- Maintain ISOL option
- Upgradable to multiuser simultaneous operation of light/heavy ions with addition of a light-ion injector

The Science of FRIB is Endorsed by NSAC and NRC



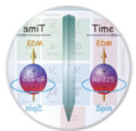
Properties of nuclei

- Develop a predictive model of nuclei and their interactions
- Understand the nuclear force in terms of QCD
- Many-body quantum problem: intellectual overlap to mesoscopic science, quantum dots, atomic clusters, etc.



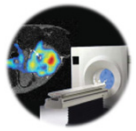
Astrophysical processes

- Chemical history of the universe
- Model explosive environments
- Properties of neutron stars, EOS of asymmetric nuclear matter



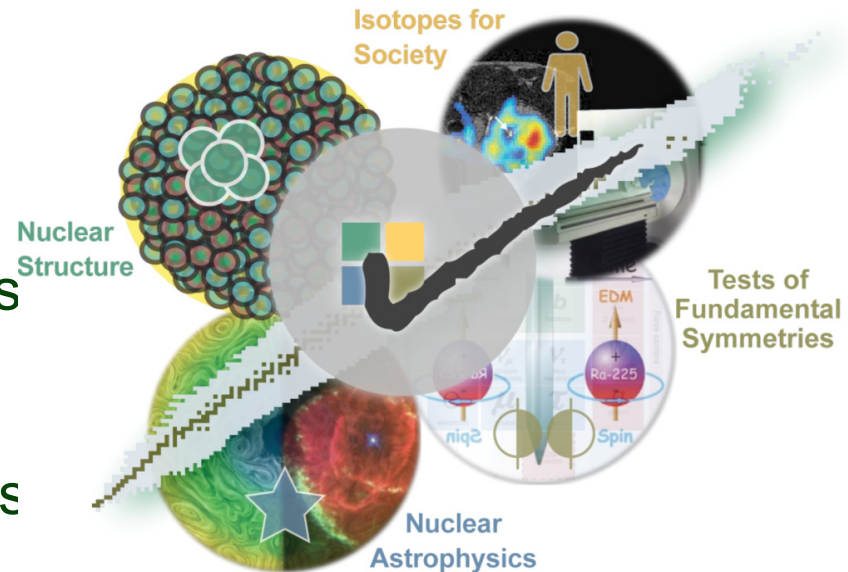
Tests of fundamental symmetries

- Effects of symmetry violations are amplified in certain nuclei



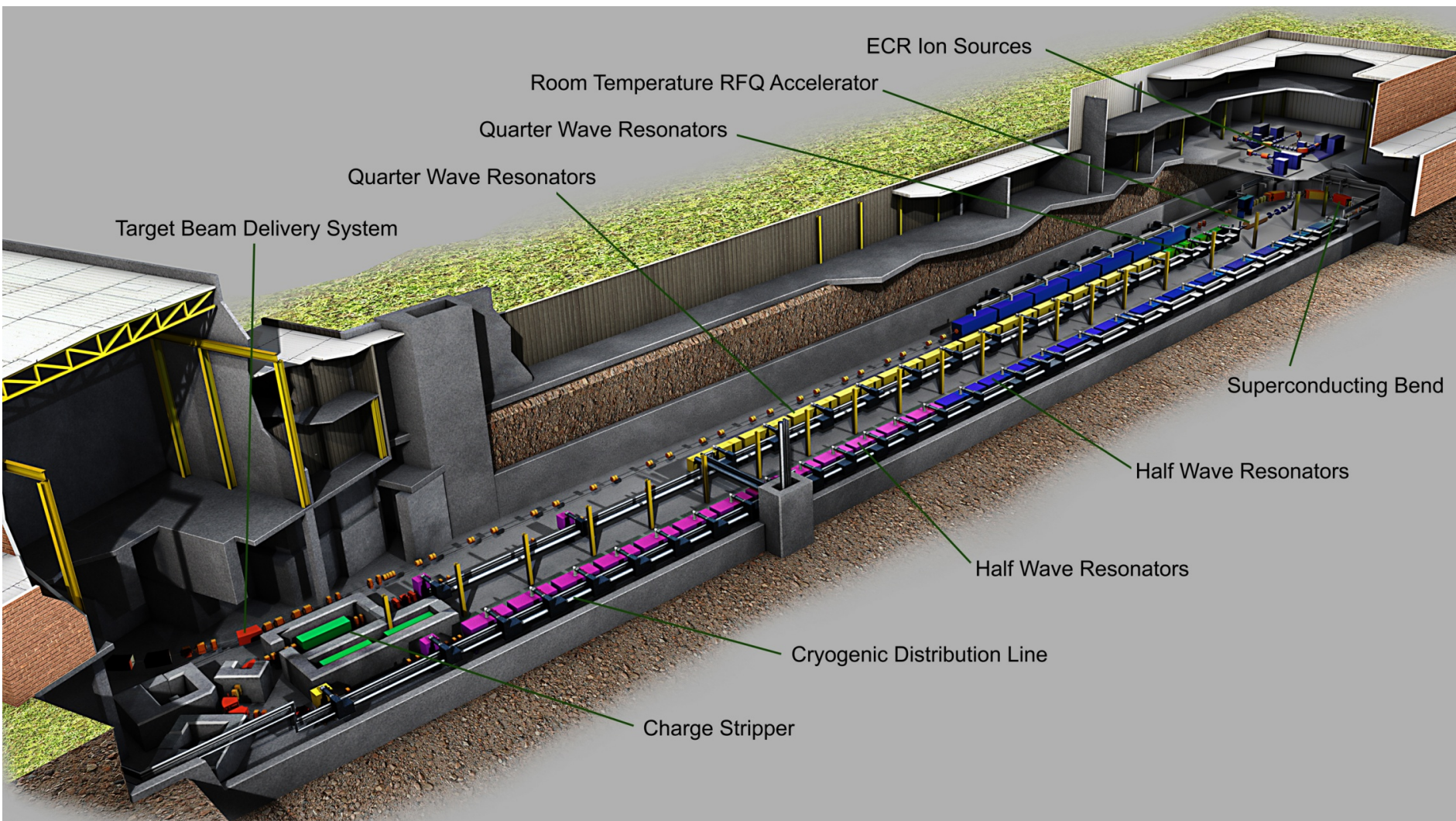
Societal applications and benefits

- Bio-medicine, energy, material sciences, national security



FRIB Driver Linac Layout

Civil Design Complete and Ready for Construction



Accelerator Design Philosophy

- A full-energy linac driver to provide beam quality that user desires
 - Full-scale CW linac using superconducting RF low- β cavities
- Meet stringent requirements demanded by experimental programs
 - Up to 400 kW of beams are focused to a diameter of 1 mm (90%)
 - Energy spread of 1% (95% peak-to-peak), and bunch length of < 3 ns
 - Intensity range of 10^8 – diagnostics & controls requirements
- Support FRIB as a national scientific user facility
 - Availability
 - Maintainability
 - Reliability
 - Tunability
 - Upgradability

Accelerator Availability & Upgradability

Design Supports Multiple Operational Scenarios

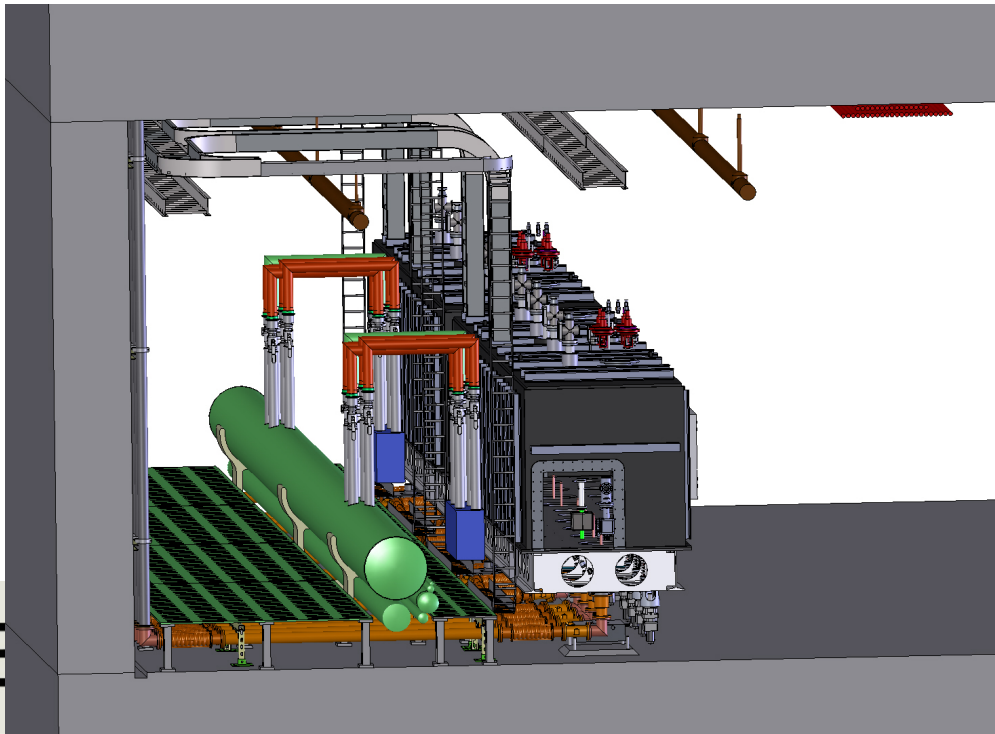
- Baseline scenario (200 MeV/u, 400 kW) with liquid Li stripper for U^{78+}
 - Multiple ion sources for enhanced availability
- Alternative scenario with He gas stripper for U^{71+}
 - Folding segment optics accommodates both stripping scenarios
- Fault scenario tolerated – comparable to SNS day-1 condition
 - Tolerate 20% cavity underperformance; single cryomodule failure; lower stripping efficiency (charge state down to U^{63+})
- Upgrade scenarios to 300 and 400 MeV/u supported

^{238}U beam

Scenario	Charge state (average)	Energy [MeV/u] (baseline)	Energy [MeV/u] (baseline + + 3 C.M.)	Energy [MeV/u] (baseline + + 12 C.M.)	Energy [MeV/u] (baseline + 12 C.M.) (35% gradient enh. for $\beta=0.29$ & 0.53)
Proposed Baseline	78+	202	228	306	413
Alternative	71+	179	202	275	375
Fault	63+	155	176	247	342

Accelerator Maintainability

- Limit uncontrolled beam loss below 1 W/m for all ion species
 - Proton: activation below 1 mSv/h; ^{238}U : machine protection & cryo load req.
- Ion sources placed at grade level 10 m above tunnel
- Adopt bayonet/U-tube and integrate heat exchanger/JT valve to cryomodule improving individual cryomodule maintainability
- Interchangeable cryomodules sacrificing peak acceleration gradient



Schematic layout of a cryomodule along with auxiliary systems in the FRIB tunnel

Accelerator Reliability

Machine Protection on Acute and Chronic Beam Losses

- Addressing key reliability and availability aspects
 - Prevent permanent accelerator component damage
 - Minimize beam loss and residual activations
 - Reduce long and frequent beam interruption, e.g. solenoid quench, cryogenic load raise
- Beam loss detection is challenging: gas monitor insensitive to low energy H⁻ beam; signal crosstalk due to folding linac footprint
 - Damages to accelerator components may occur in 40 μ s; MPS budget is 35 μ s: 15 μ s diagnostic, 10 μ s control, and 10 μ s beams in pipe
- Machine protection on acute (fast) chronic (slow) beam loss
 - Monitor beam loss using halo monitor rings at warm region between cryomodules
 - Gas chamber detection on acute beam loss at high energy
 - Beam current monitor at entrance and exit of each linac segments

SHV-20
FEEDTHRU

2 ¾ INCH
CONFLATS

NIOBIUM
HALO RING

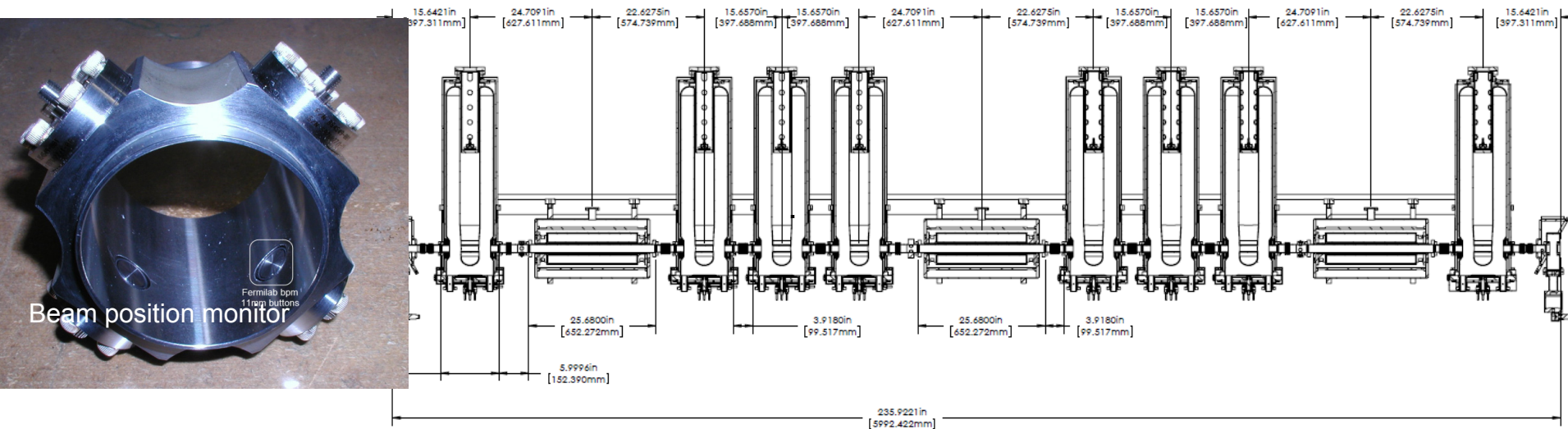
Schematic drawings of the halo
monitor rings for beam loss detection

Accelerator Tunability

Example: Linac Segment 1 Cold BPM for On-line Tuning

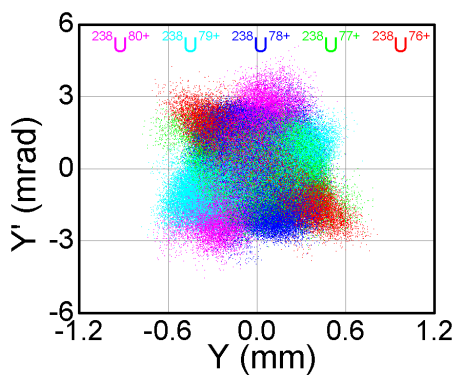
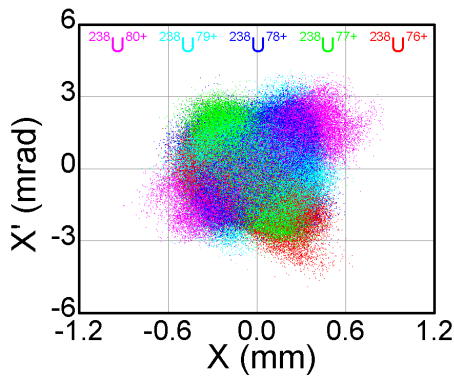
- Implement 39 “cold” beam position monitors (BPM) to QWR segment – allowing practical beam steering for increased accelerator availability
 - Cold BPM facilitates response-matrix-based on-line tuning for greatly improved machine availability
 - Mitigated prior design issue of large transverse phase advance (π) between BPMs (warm)

Schematic drawing of a QWR cryomodule containing 8 cavities, 3 solenoids and 3 cold BPMs



Accelerator Physics Challenges

- Combined challenges of heavy-ion & high-power accelerator
 - Fractional uncontrolled beam loss at 10^{-6} per m level
 - Protons: activation & shielding issues; ^{238}U : material damage & heat load
- Limited aperture of accelerating structures (low- β vs. elliptical cavity)
- Simultaneous acceleration & overlapping of multi-charge-state beams
 - Achromatic optics design (transverse) & cavity placement (longitudinal)
 - Diagnostics and control capabilities
- Accurate alignment of “cold” elements in cryomodules
 - 9-T solenoid & BPM to be aligned to < 1 mm under cryogenic condition
- Stringent beam-on-target requirements
 - Requiring corresponding beam diagnostics & control



Transverse phase space with 5 charge states of uranium particles overlapping

Technology Challenges

▪ Charge stripping

- Solid foils unable to survive on the high-power, high-charge-state beams
- Collaboration with ANL on liquid-lithium stripping film (low vapor pressure)
- Collaboration with BNL on plasma window to contain helium-gas stripper
 - » Also benefit from development progress at RIKEN on helium gas stripping

▪ Superconducting RF

- > 10 years of development at MSU on low- β SRF, benefiting from seminar work at INFN
- Collaboration with JLab and ANL
 - » subcomponent development & processing
- Benefit from consultations with world experts from INFN, TRIUMF, JLab, ANL, FNAL etc.

▪ Collaboration with major institutes and laboratories world-wide on key accelerator subsystems

World-wide Partnership and Collaboration with Major Accelerator Institutes

- **ANL**
 - Liquid lithium stripper
 - Beam dynamics verification; $\beta=0.29$ HWR design; SRF coupler and tuner
 - **BNL**
 - Plasma window & charge stripper, physics modeling, database
 - **FNAL**
 - Diagnostics, SRF processing
 - **JLab**
 - Cryogenics systems design
 - QWR & HWR hydrogen degassing
 - PANSOPHY e-traveler
 - HWR processing & certification*
 - **LANL**
 - Proton ion source, RFQ
 - **LBNL**
 - ECR ion source; beam dynamics**
 - **ORNL**
 - Diagnostics, controls
 - **SLAC****
 - Cryogenics**, SRF multipacting**, physics modeling
 - **RIKEN**
 - Helium gas charge stripper
 - **TRIUMF**
 - Beam dynamics design, SRF, physics modeling
 - **INFN**
 - SRF technology
 - **KEK**
 - SRF technology, SC solenoid
 - **IMP**
 - Magnets*
 - **Budker Institute, INR Institute**
 - Diagnostics
 - **Tsinghua Univ. & CAS**
 - RFQ
 - **ESS**
 - AP*
 - **DTRA**
 - RFQ power supply**
- * Under discussion or in preparation
 ** Completed



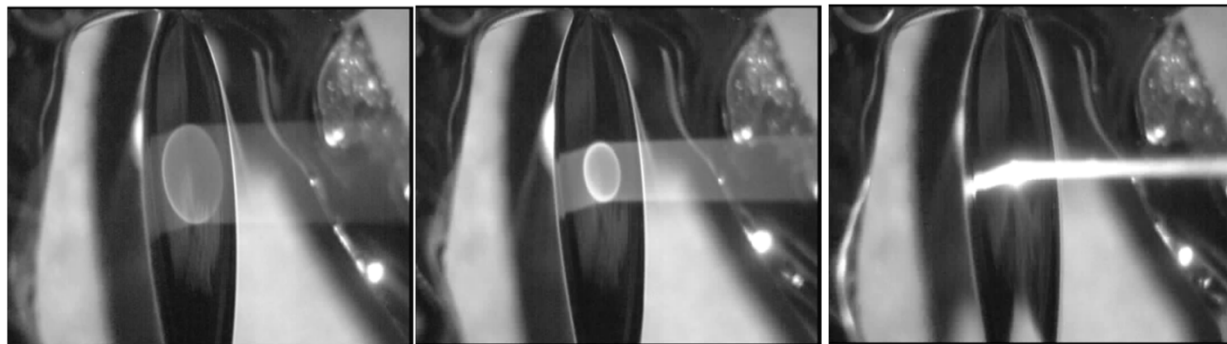
Li Charge Stripper Tested with Beam Power

Successful LANL IS Restoration at MSU & Beam Power Test at ANL

- Liquid lithium film established with controllable thickness and uniformity
- LEDA Ion Source (IS) beam commissioned at MSU
 - Beam commissioned at MSU after restoring with new cooling and power supply system after more than 10 years of storage.
- Beam power tests on liquid lithium film successfully performed at ANL
 - The film sustained ~200% of FRIB maximum power density deposition



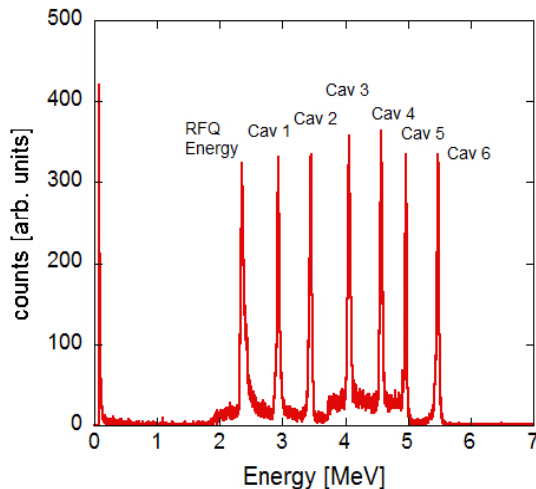
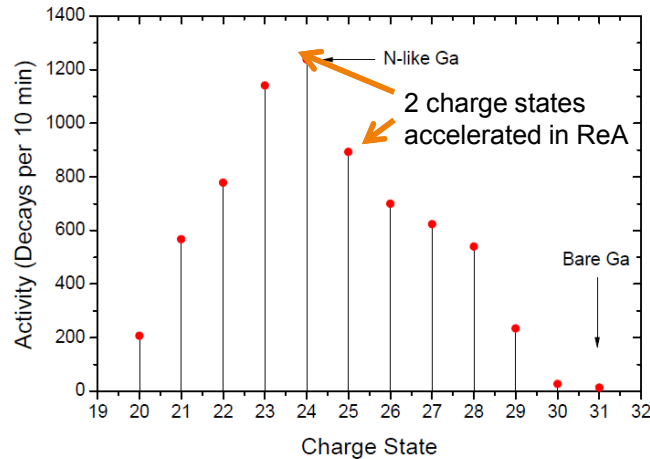
Liquid lithium film flowing at high speed (~ 50 m/s) intercepting a proton beam of about 60 kV at ANL. The test produced power deposition densities similar to the FRIB uranium beams.



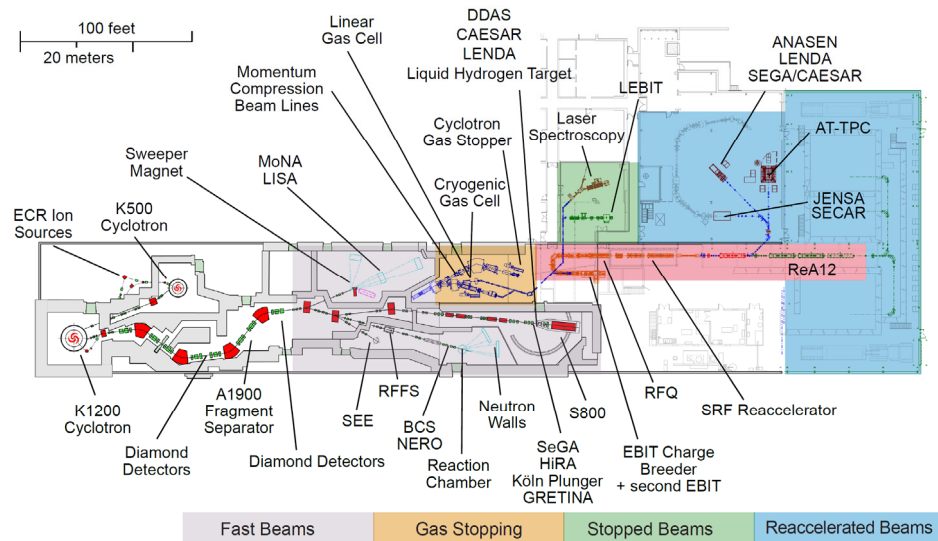
System Integration Proceeding: Rare Isotope ^{76}Ga Produced and Accelerated



EBIT charge breeding of Ga-76

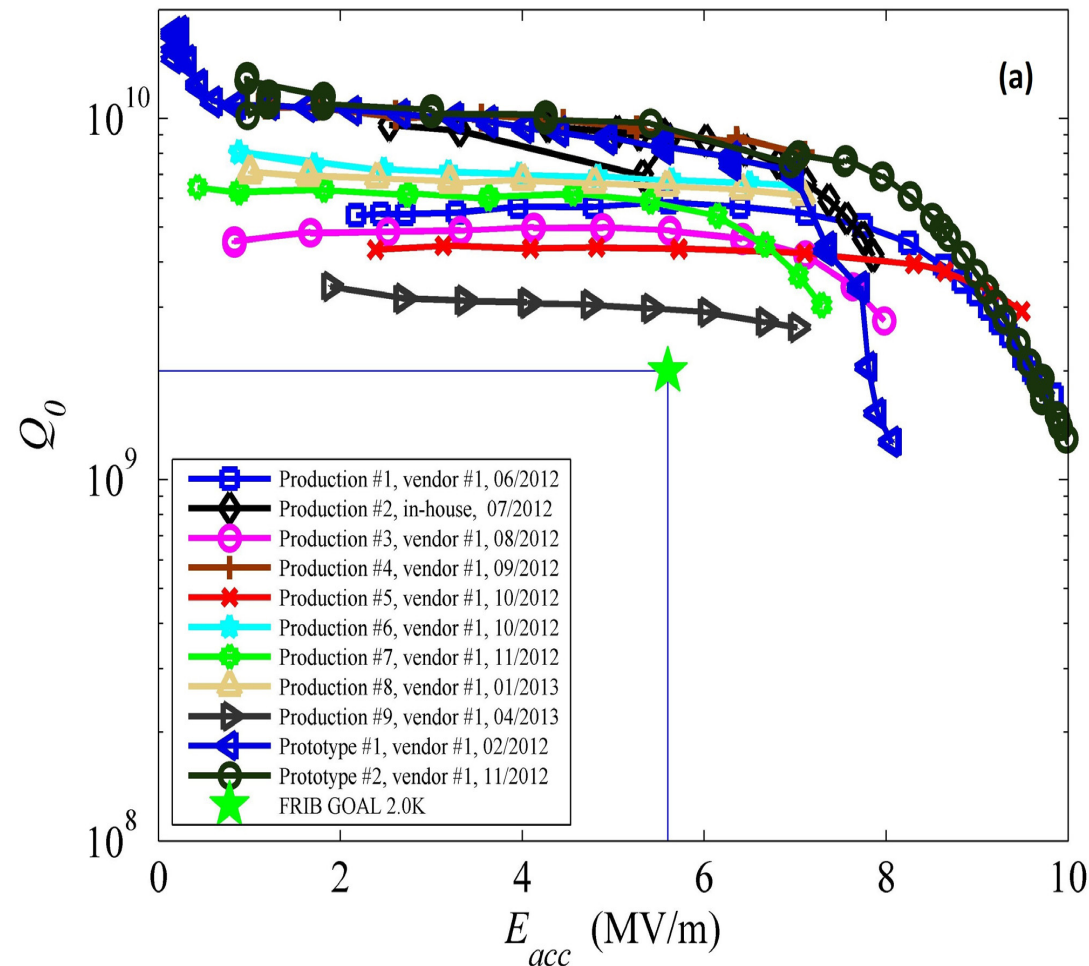


- Superconducting cyclotrons accelerate ^{76}Ge beam to 130 MeV/u; ^{76}Ga produced; stopped in the ANL gas cell; Charge Breeding in the EBIT Source
- Re-acceleration in the ReA accelerator
 - Using RFQ and $\beta=0.041$ cryomodules
- SRF technology mature for FRIB project



Successful Prototyping of $\beta_0=0.085$ QWRs

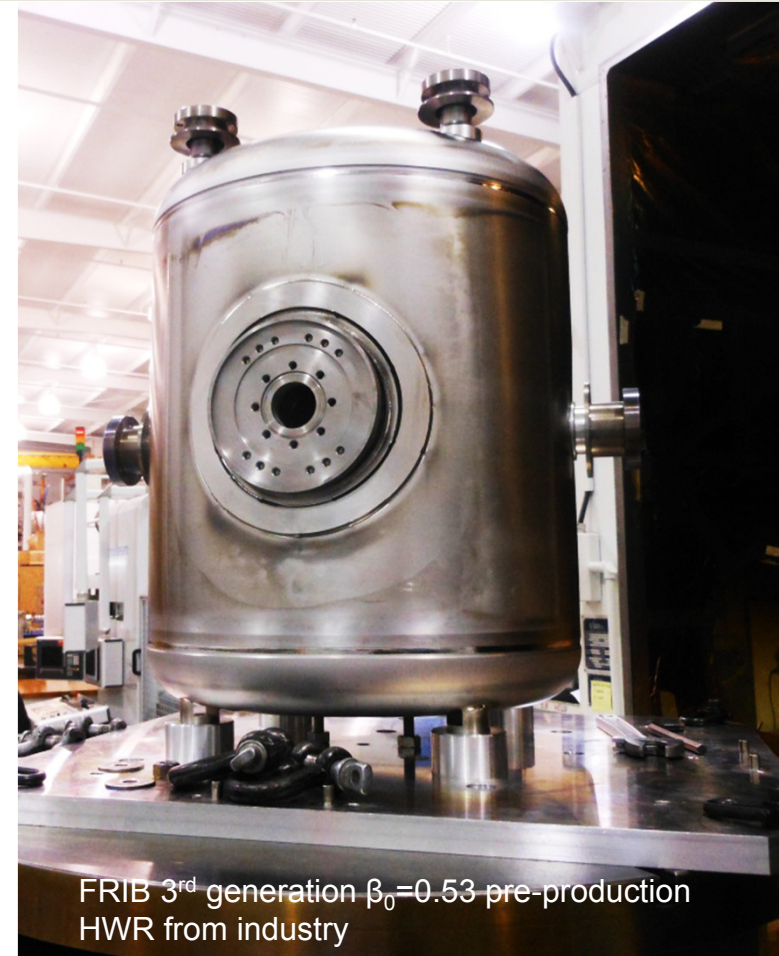
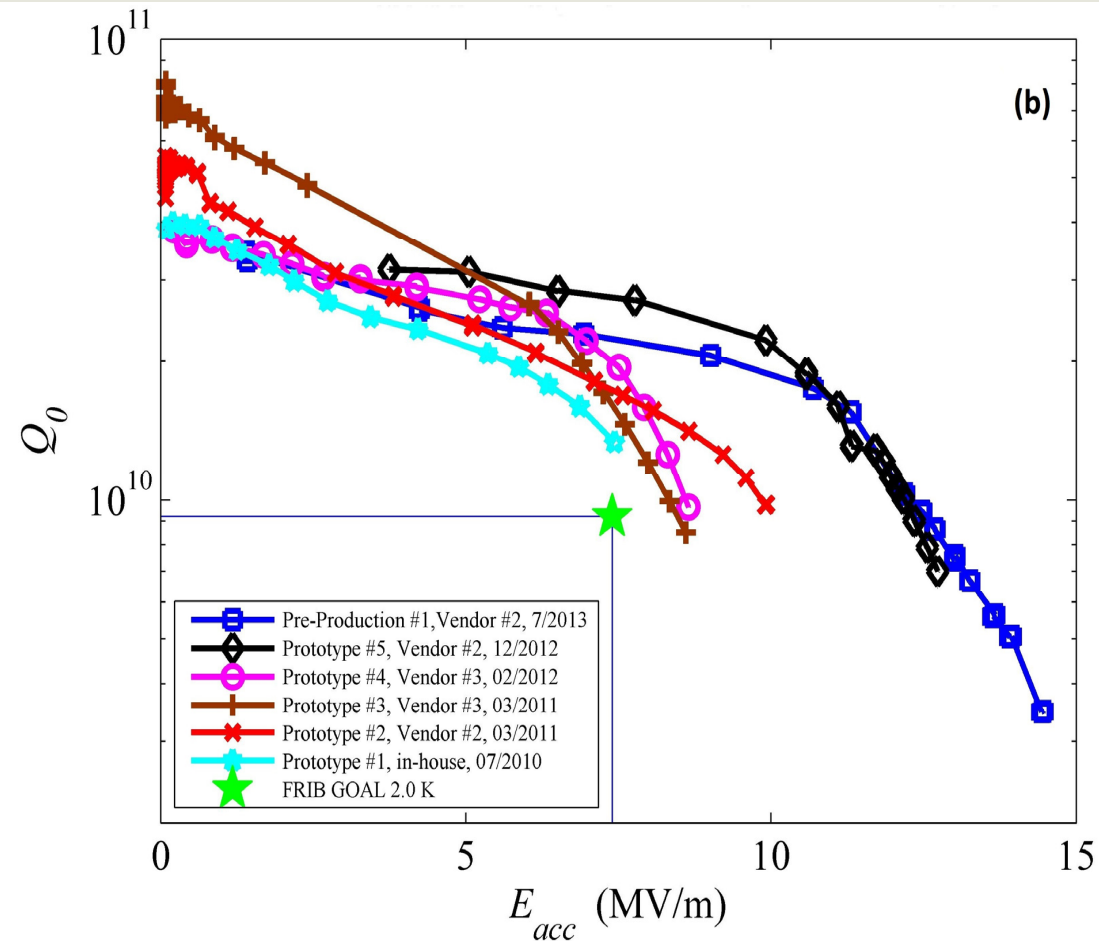
Significant Margin Allows High Machine Availability



Margin of about a factor of 3 in Q , 40% in E field



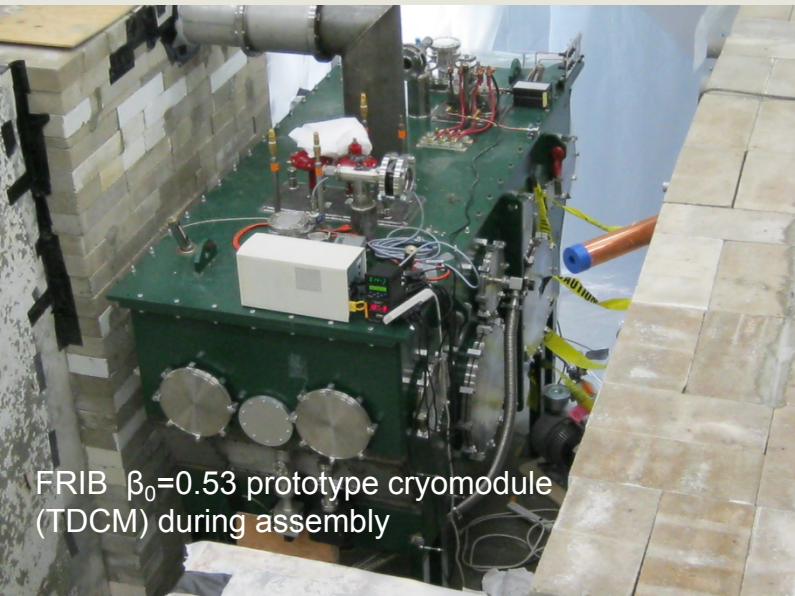
From Prototyping to Pre-production: $\beta_0=0.53$ HWR Cavities from Multiple Vendors Meet Performance Goals



Industrial provider contracted for mass production successfully delivers cavities

$\beta_0=0.53$ Prototype Cryomodule Tests

Successfully Meeting R&D Milestones in SRF Tests



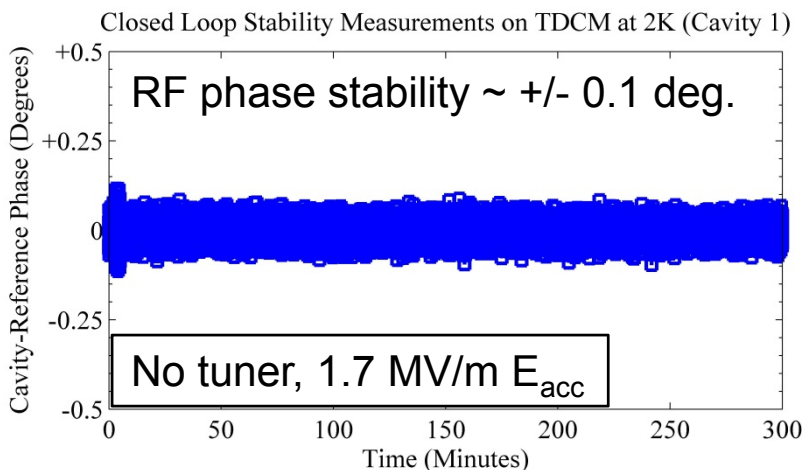
FRIB $\beta_0=0.53$ prototype cryomodule (TDCM) during assembly

■ FRIB Technology Demonstration Cryomodule R&D milestones completed

- TDCM operates stably at 2 K temperature with excellent cryogenic stability
- Cavities continually locked to design frequency; excellent low-level RF control
- Coupler operated at full CW power (4.5 kW) in full reflection within specified cryogenic load
- Magnetic shielding efficiency demonstrated
- Ancillary components (cavity, low-level control, coupler, tuner) operating

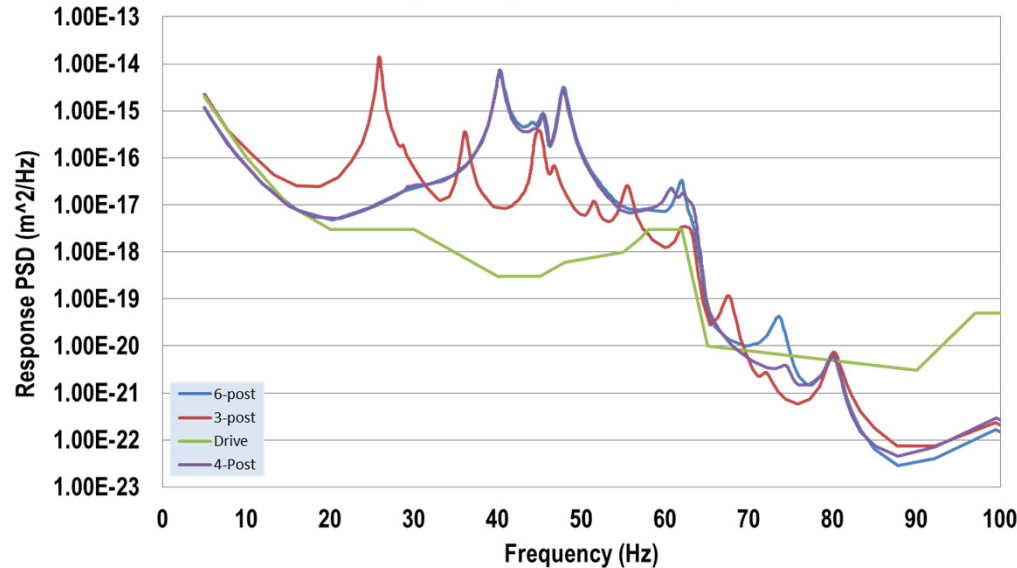
■ Lessons learned to benefit the design of FRIB preproduction cryomodules

- Team coordination, engineering culture enforcement, magnetic material management, tuner noise, coupler/cavity multipacting, solenoid lead heat load/pressure drop, NSCL cryogenics issues

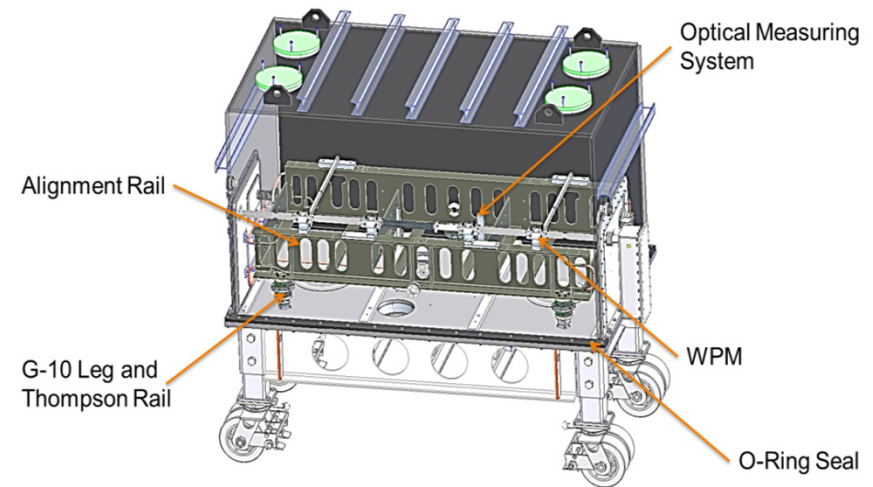


Bottom-Up Cryomodule Design Validated with High Alignment Accuracy (~ 0.1 mm)

Response Spectrum Comparison

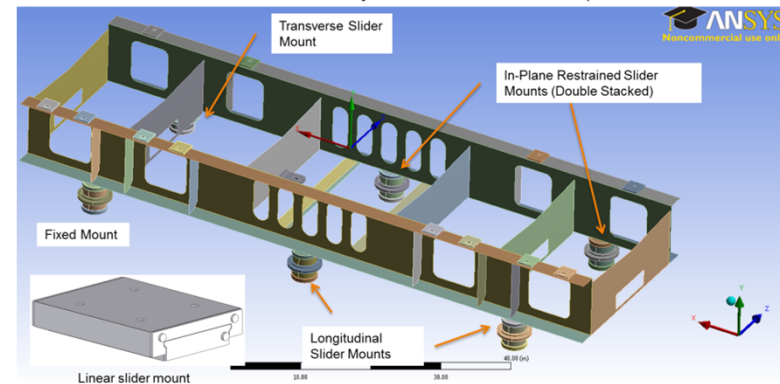


FRIB Engineering Test Cryomodule schematics



- Engineering Test Cryomodule (ETCM) successfully verified alignment during liquid nitrogen cool-down & vacuum vessel enclosure
- 4-point support established as design choice based on vibrational response analysis
- Vibration evaluation and test underway with actual load and realistic configuration

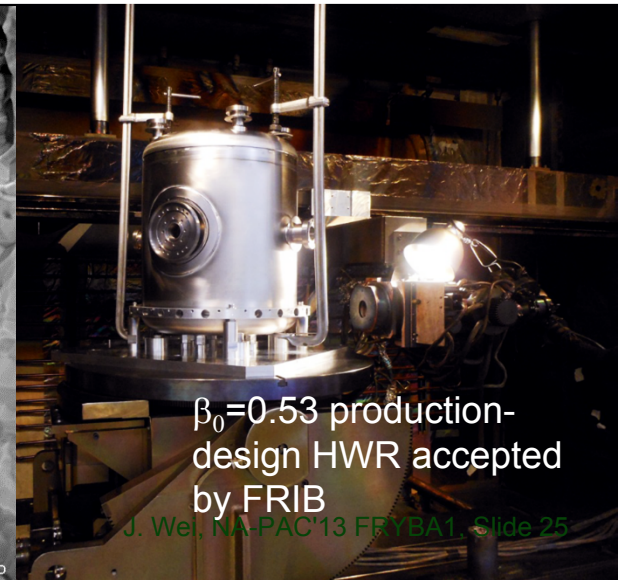
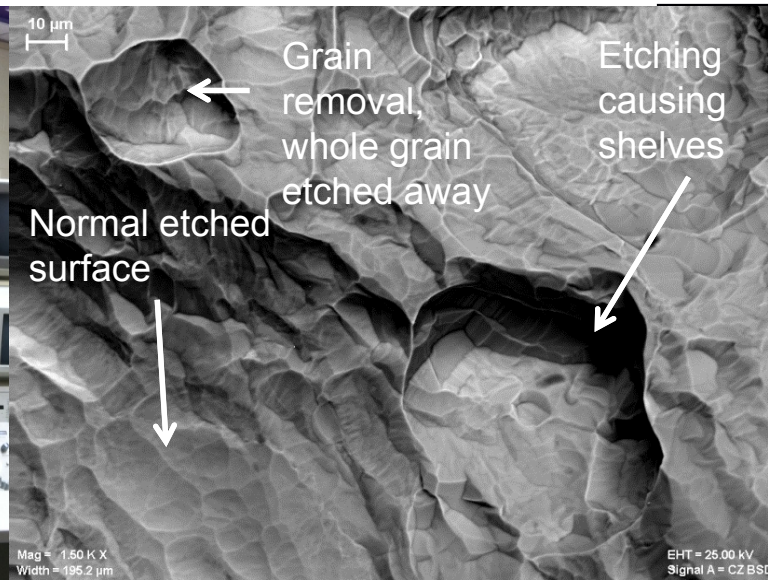
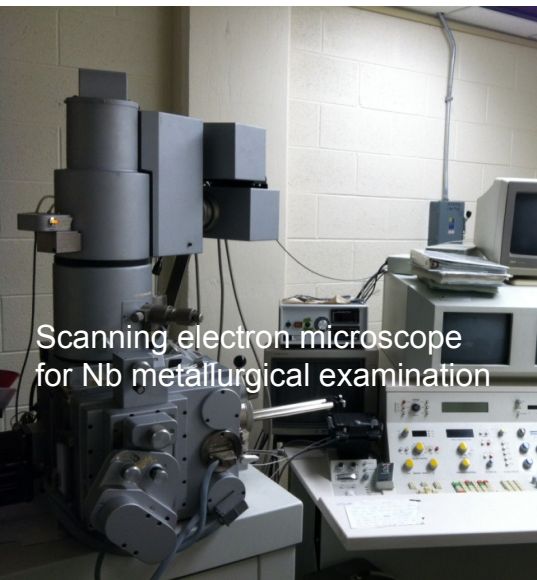
ANSYS Model for the Center Rail Assy allows either 3 or 4 or 6 post simulations



SRF Acquisition Strategy Established

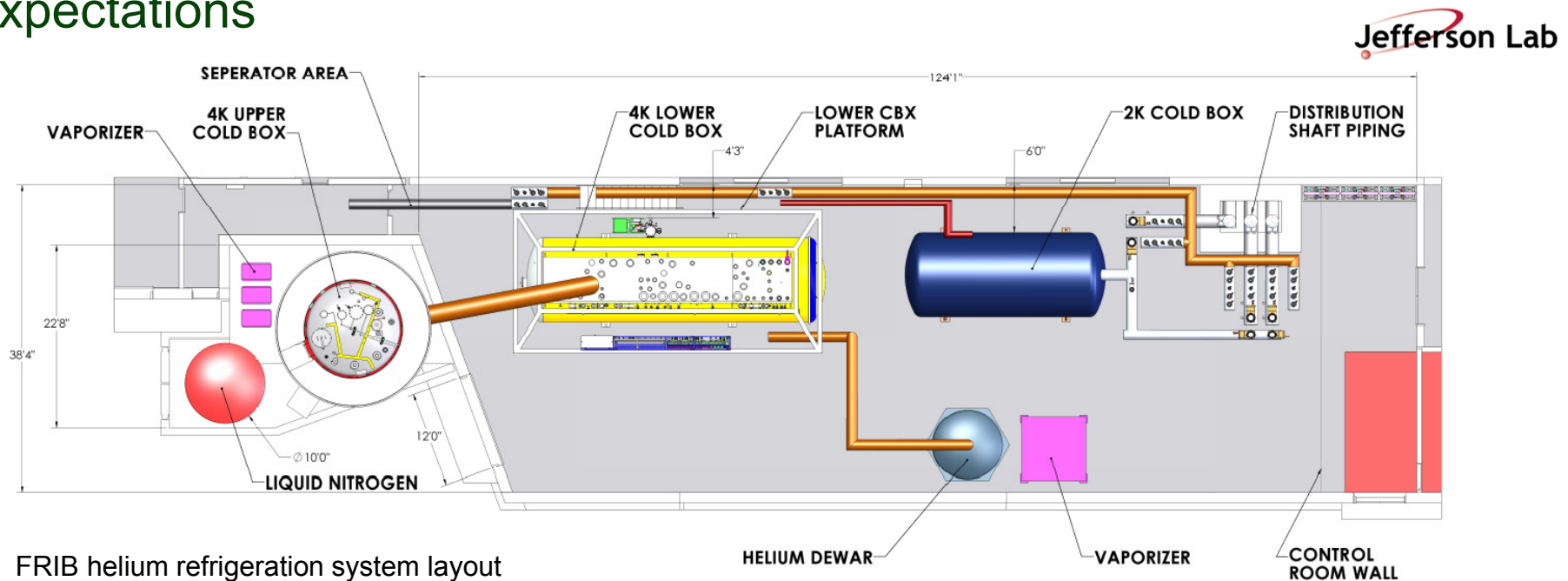
Active Vendor Engagement is Key to Procurement Success

- FRIB actively expanding vendor base; risk sharing to reduce cost
 - Material (Nb, NbTi) order at final contracting stage with 3 providers
 - Largest SRF cavity order to a domestic provider: 174 $\beta_0=0.53$ HWRs
 - » 3 vendors delivered FRIB cavities; 7 vendors responded to recent HWR request-for-proposal (RFP)
 - » 174 cavities awarded in 3 production steps of 2-cavity, 10-cavity, and mass production (with 10% excess)
 - $\beta_0=0.29$ pre-production HWR: awarded to industry
 - $\beta_0=0.085$ pre-production QWR: awarded to industry
- Monitoring and acceptance procedures established to assure quality



Cryoplant 4.5 K Cold Box Under Contract

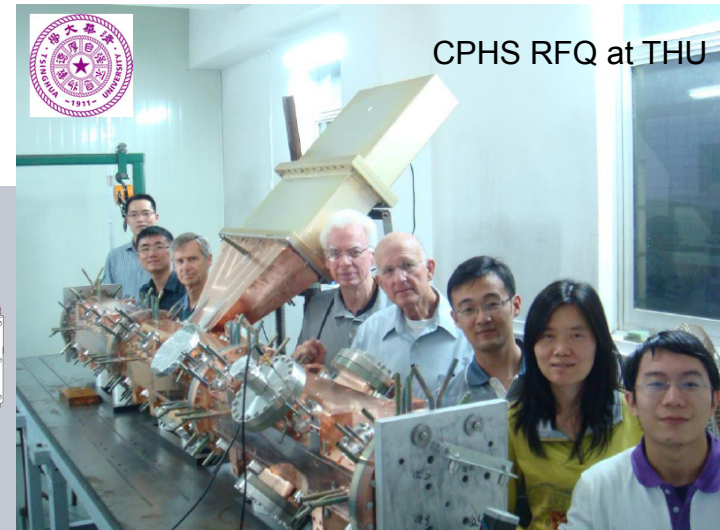
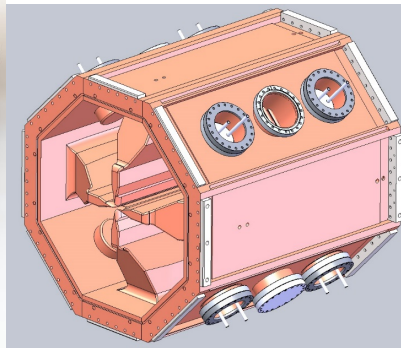
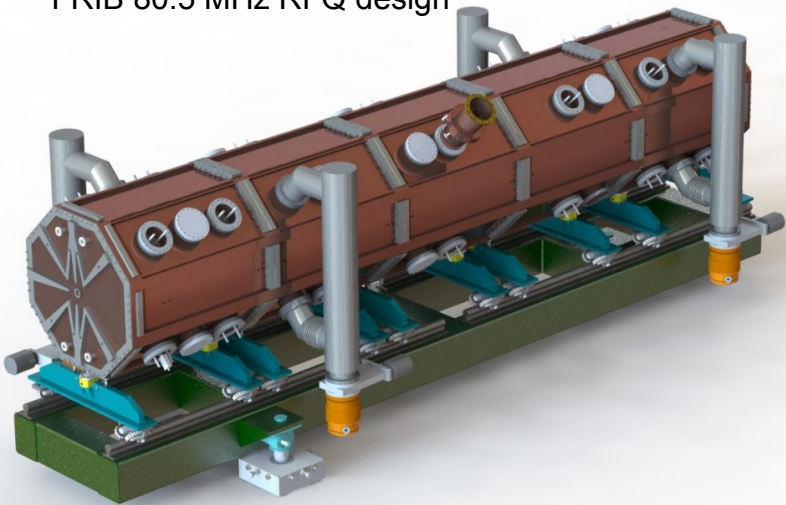
- FRIB and JLab agreed on Work For Others (WFO) for FRIB cryoplant design/acquisition to complete by FY2017
 - MSU responsible for procurements, installation support and controls – in full support of JLab team's needs
- 4.5 K cold box contracted, meeting project cost and schedule expectations



RFQ Under Contract

- Design-build contract awarded to industry
- RFQ world experts ensuring solid design and engineering
- Collaborating with THU for design/engineering coordination and quality assurance

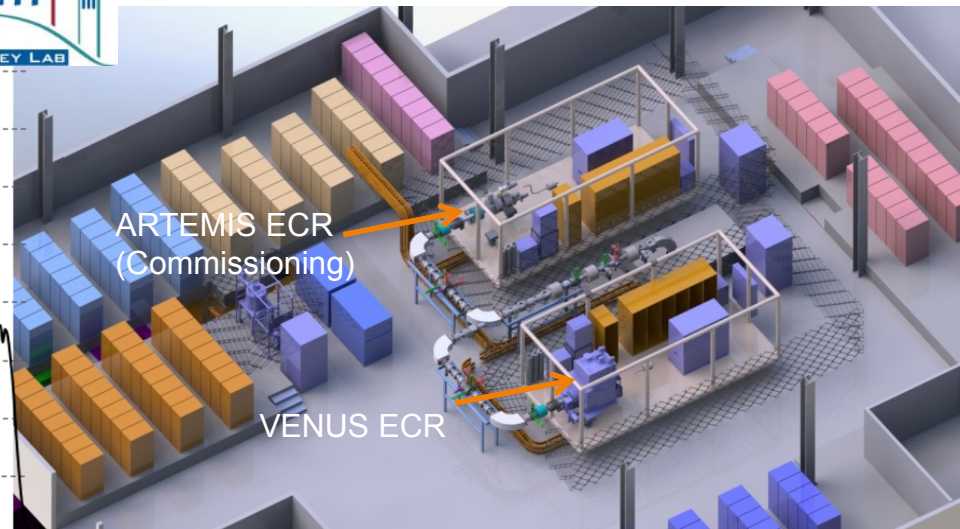
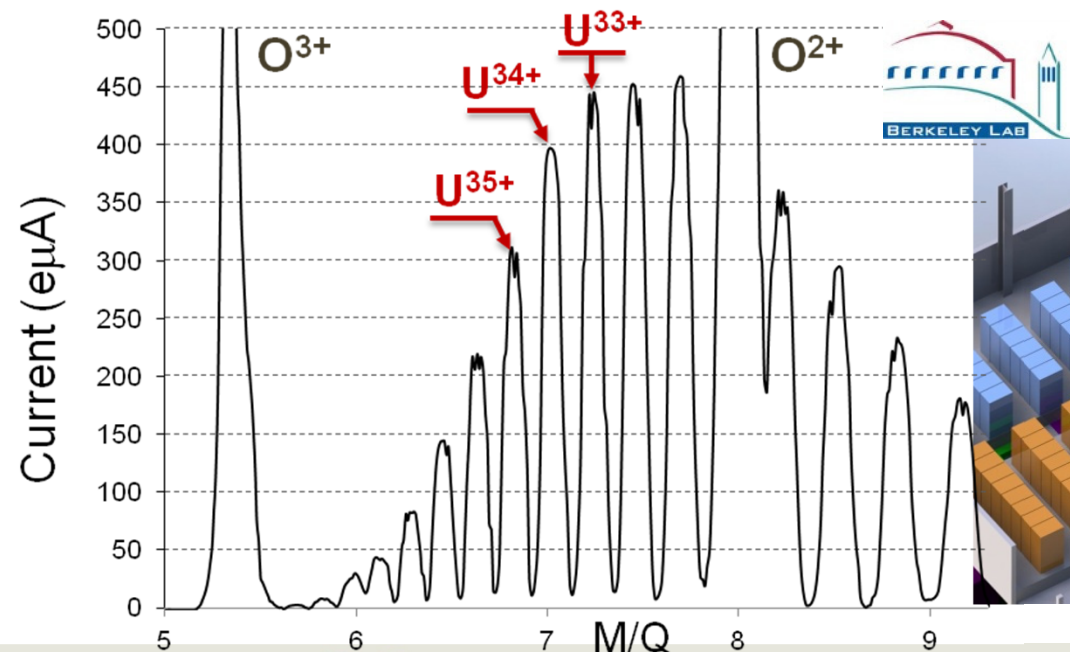
FRIB 80.5 MHz RFQ design



Early Front End Establishment Planned

Ion Sources at LBNL and MSU Demonstrated Performance

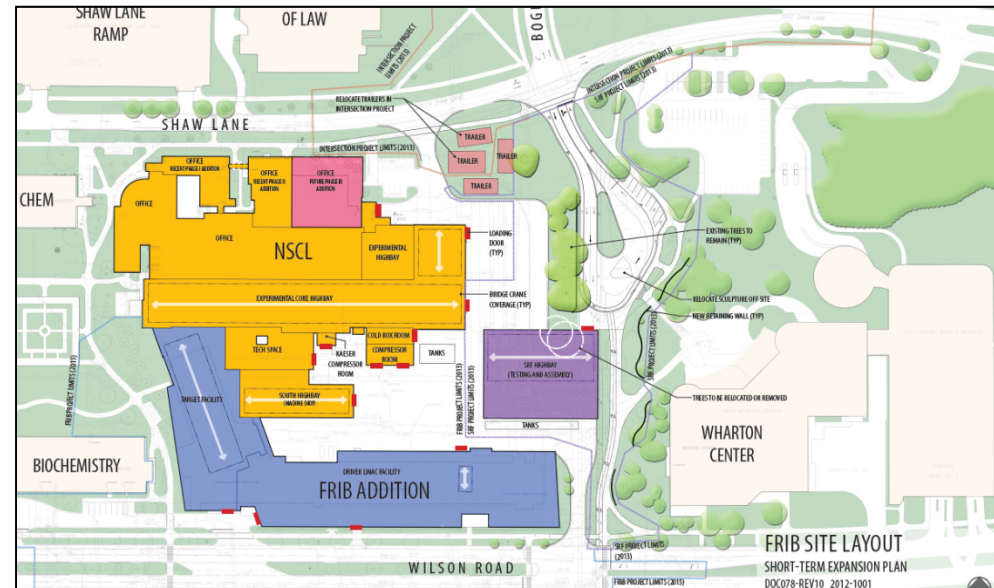
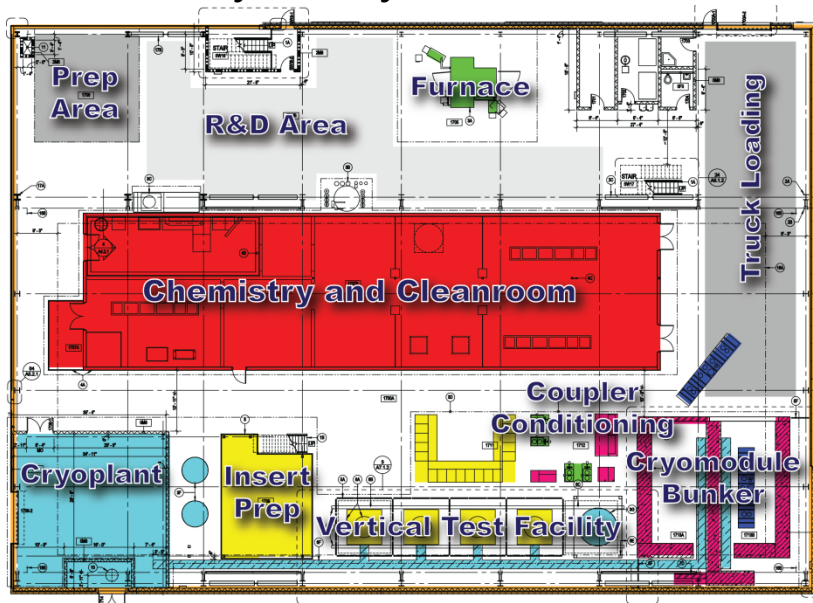
- Two ECRs to cover FRIB Project's commissioning & operation needs
 - ECRIS based on ARTEMIS design running at MSU is to be tested in FRIB configuration in 2012; adequate for commissioning & light ion operations
 - High-performance superconducting ECRIS is based on VENUS design; LBNL/VENUS source delivered twice FRIB ^{238}U required intensity in 2011
 - » Team and expertise established at MSU
 - » Fabrication of ECR sextupole/solenoid coils in collaboration with LBNL



MSU-Supported SRF Infrastructure

SRF Furnace Commissioned; SRF High Bay in Construction

- Heat treatment furnace commissioned
- MSU funded SRF high bay (~2500 m²) in construction
 - 2 cryomodule test caves; 4 vertical test dewars; expanded cleanroom & chemistry facility

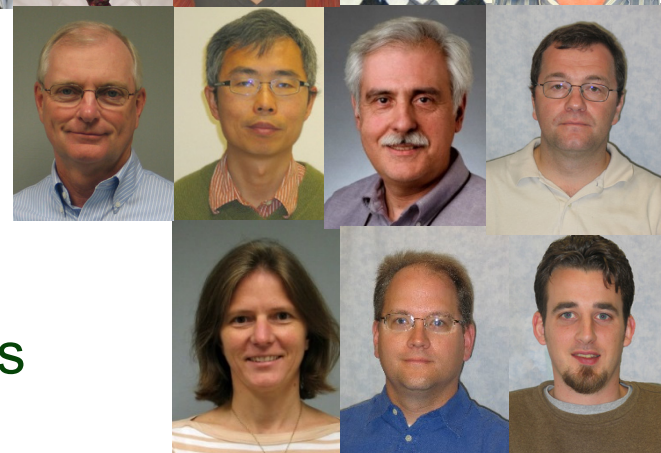


SRF High Bay under construction at MSU



Attracted World Experts in Key Areas

Aggressively Strengthened Technical Team Leadership

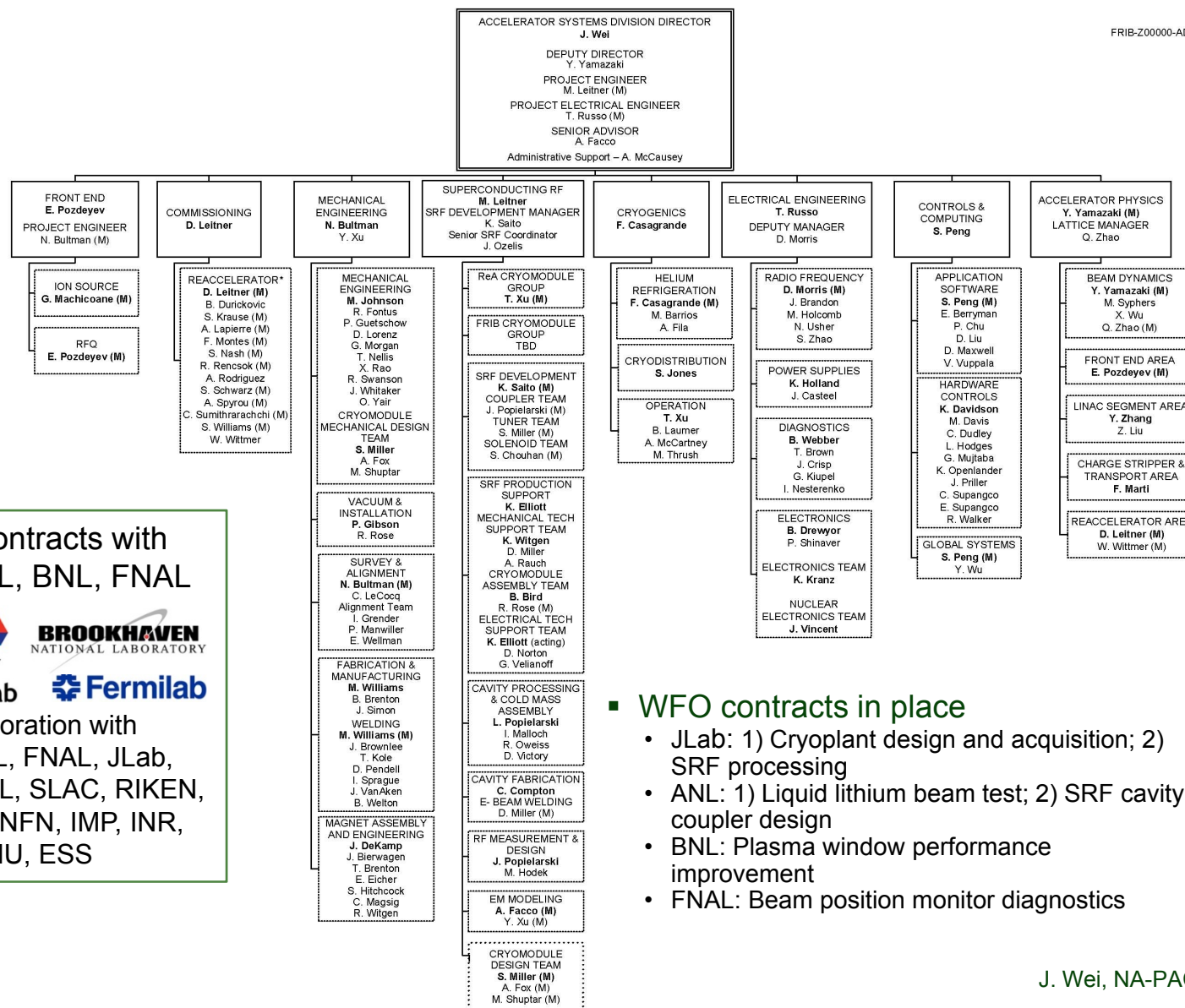


- Recruited top SRF scientists and pioneers
 - A. Facco (INFN), K. Saito (KEK)
- Built cryogenics team to leverage SNS success
 - F. Casagrande (ORNL), T. Xu (ORNL)
- Grouping of seasoned leaders in accelerator physics and engineering
 - N. Bultman (LANL), K. Holland (NSCL), D. Leitner (LBNL), M. Leitner (LBNL), S. Lidia (LBNL), F. Marti (NSCL), D. Morris (NSCL), S. Peng (SLAC), E. Pozdeyev (BNL), T. Russo (BNL), Y. Yamazaki (J-PARC), B. Webber (FNAL), J. Wei (BNL), Y. Zhang (ORNL)

ASD Organization Ready for Scope Delivery

Area and Control Account Managers Integrating Scope

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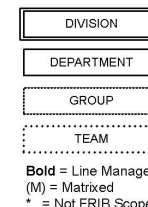
WFO contracts with
JLab, ANL, BNL, FNAL



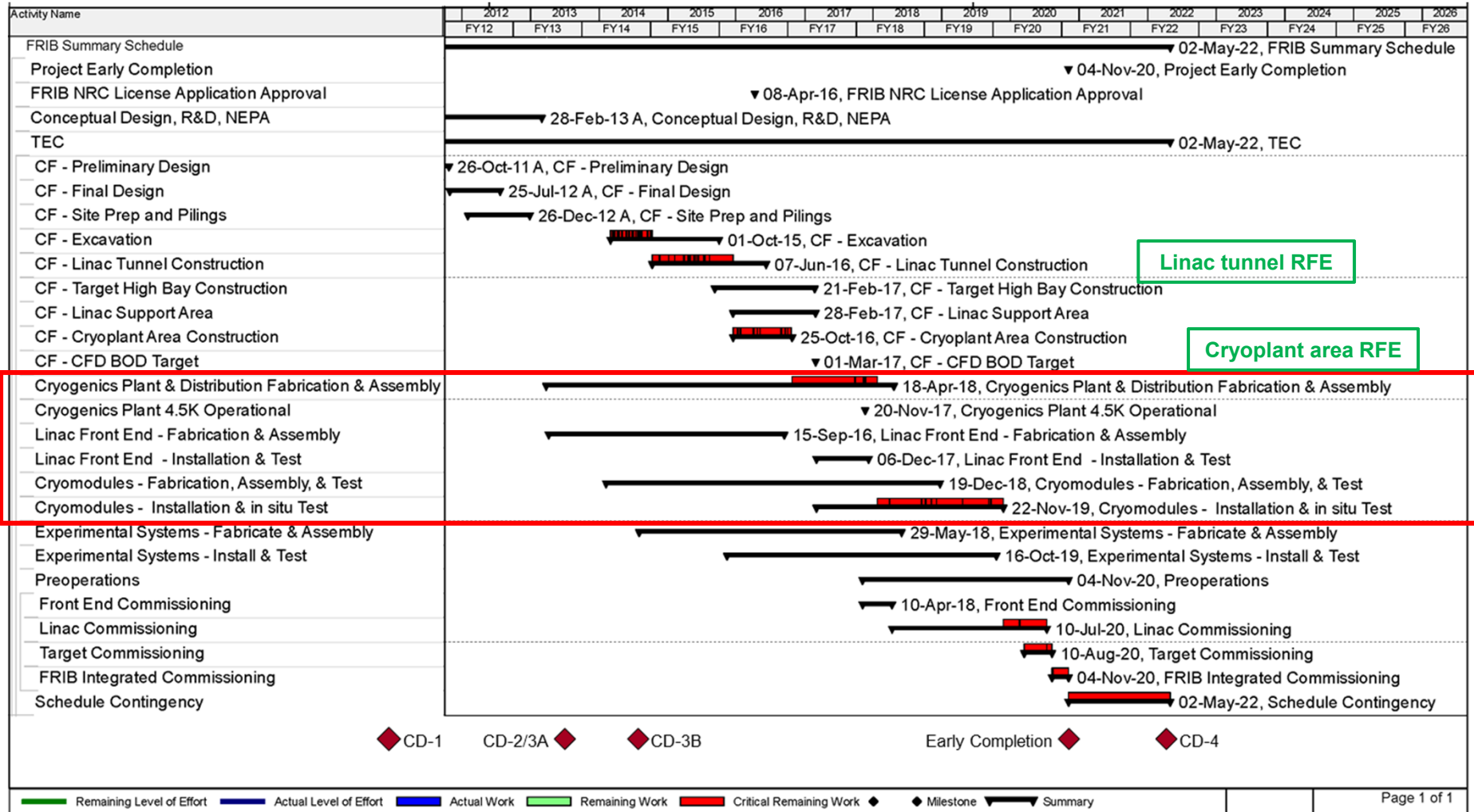
Collaboration with
ANL, BNL, FNAL, JLab,
LBNL, ORNL, SLAC, RIKEN,
TRIUMF, INFN, IMP, INR,
THU, ESS

WFO contracts in place

- JLab: 1) Cryoplant design and acquisition; 2) SRF processing
- ANL: 1) Liquid lithium beam test; 2) SRF cavity coupler design
- BNL: Plasma window performance improvement
- FNAL: Beam position monitor diagnostics



Integrated Project Schedule



Summary

- FRIB project is proceeding with scope, schedule and cost baselined and ready for civil construction start
- Accelerator design meets FRIB performance requirements
 - Accelerator lattice footprint frozen since June 2011
 - Detailed developments continue on critical components- charge stripper and SRF
 - System designs optimized and value engineered for availability, maintainability, reliability, tunability, and upgradability
- Acquisition strategy has been meeting performance, cost & schedule requirements
- An excellent team is in place to lead accelerator systems delivery
- FRIB is looking for dedicated fellows & seasoned colleagues to join the project, and also welcomes collaboration in all forms
- Thank you!

Coauthors

■ J. Wei^{#1}, D. Arenius², N. Bultman¹, F. Casagrande¹, C. Compton¹, K. Davidson¹, J. DeKamp¹, B. Drewyor¹, K. Elliott¹, A. Facco^{1,4}, V. Ganni², P. Gibson¹, T. Glasmacher¹, K. Holland¹, M. Johnson¹, S. Jones¹, R.E. Laxdal⁵, D. Leitner¹, M. Leitner¹, G. Machicoane¹, F. Marti¹, D. Morris¹, J. Nolen^{1,3}, J. Ozelis¹, S. Peng¹, J. Popielarski¹, L. Popielarski¹, E. Pozdeyev¹, T. Russo¹, K. Saito¹, J. Savino¹, R. Webber¹, M. Williams¹, T. Xu¹, Y. Yamazaki¹, A. Zeller¹, Y. Zhang¹, Q. Zhao¹

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- ⁵ TRIUMF, Vancouver, Canada

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